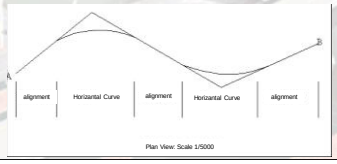
COURSE SYLLABUS

Weekly Subjects and Related Preparation Studies

WEEK	SUBJECT	DATE	
1	INTRODUCTION	05.10.2020	Monday
	BASIC DEFINITIONS (TERMS)	08.10.2020	Thursday
2	ROAD USER CHARECTERISTICS	12.10.2020	Monday
	VEHICLE MOVEMENTS	15.10.2020	Thursday
3	TRAFFIC FLOW AND CAPACITY	19.10.2020	Monday
	ROADWAY CASSIFICATION AND GEOMETRIC STANDARDS	22.10.2020	Thursday
4	HIGHWAY ROUTE STUDY	26.10.2020	Monday
	NATIONAL HOLIDAY	29.10.2020	Thursday
5	HORIZANTAL CURVE AND SUPERELEVATION (Project sheet distribution)	02.11.2020	Monday
	HORIZANTAL CURVE AND SUPERELEVATION	05.11.2020	Thursday
6	**APPLICATION (VEHICLE MOVEMENTS)**	09.11.2020	Monday
	ASSIGNMENT - APPLICATION (TRAFFIC)	12.11.2020	Thursday
7	**APPLICATION (HORIZANTAL CURVE)**	16.11.2020	Monday
	ASSIGNMENT-APPLICATION (HORIZANTAL CURVE)	19.11.2020	Thursday
8	**MID TERM EXAM**	23.11.2020	Monday
		26.11.2020	Thursday
9	TRANSITION CURVES- ASSIGNMENT (HAND IN)	30.11.2020	Monday
	TRANSITION CURVES – ASSIGNMENT (RETURN)	03.12.2020	Thursday
10	VERTICAL ALIGNMENT AND VERTICAL CURVE	07.12.2020	Monday
	VERTICAL ALIGNMENT AND VERTICAL CURVE	10.12.2020	Thursday
11	**APPLICATION (TRANSITION CURVES)**	14.12.2020	Monday
	ASSIGNMENT	17.12.2020	Thursday
12	**APPLICATION (VERTICAL CURVE)**	21.12.2020	Monday
	APPLICATION (VERTICAL AND HORIZANTAL CURVE)	24.12.2020	Thursday
13	PLANNING OF INTERSECTION AND URBAN ROADS	28.12.2020	Monday
	PROJECTS DELIVERY DUE DATE (December 30, 2020)	31.12.2020	Thursday
14	DRAINAGE IN HIGHWAY	04.01.2020	Monday
	MAKEUP EXAM	07.01.2020	Thursday

BASIC CONCEPTS

Highway, road: A main public road, especially one connecting towns and cities; an open way (generally public) for travel or transportation.
Highway traffic: The movement of vehicles, people, etc., on highway for a particular purpose.
Location, route: The prescribed course to be travelled from a specific point of origin to a specific destination.
Plan: A drawing or diagram drawn on a plane; the projection of a road on a topographical map; scale drawing of a structure.
Center line: A real or imaginary line that is equidistant from the surface or sides of something (road).
Alignment: The position or the layout of the central line of the highway.
Horizontal curve: A horizontal curve provides a transition between two tangent strips of roadway, allowing a vehicle to negotiate a turn at a gradual rate rather than a sharp cut.

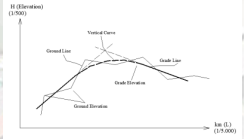


BASIC CONCEPTS

Profile: A vertical section of the surface of the ground, or of underlying strata, or both, along any fixed line. On a highway, the profile is usually taken along the centre line.



Vertical curve: Curve that is applied to make a smooth and safe transition between two grades on a roadway or a highway.



Ground line: The line representing the elevation of ground level.
Ground elevation: The elevation of a point on the ground level; in other words, the elevation of a point on the ground line.

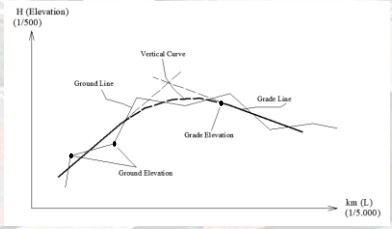
BASIC CONCEPTS

Ground line: The line representing the elevation of ground level.

Ground elevation: The elevation of a point on the ground level; in other words, the elevation of a point on the ground line.

Grade line: A longitudinal reference line or slope to which a highway or railway is built.

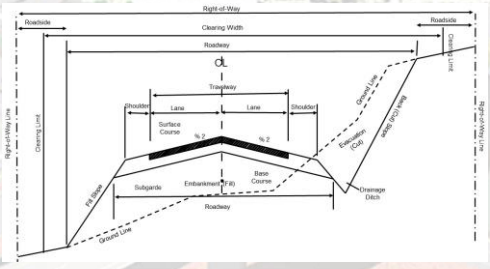
Grade elevation: The elevation of a point on the profile of a built road; in other words, the elevation of a point on the grade line.



BASIC CONCEPTS

Cross-section: A vertical section of the road axis.

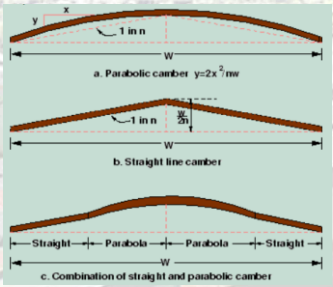
Typical cross-section: A cross-section which provides the standart measures and characteristics of a road to be consructed. On a typical cross-section the dimensions of the platform, pavement, lane, shoulder, cut ditch and refuge; the cut and fill slopes; the layers and the materails of the pavement and the width of right-of-way are shown.



BASIC CONCEPTS

Camber (crown)

Camber or cant is the cross slope provided to raise middle of the road surface in the transverse direction to drain off rain water from road surface. The common types of camber are parabolic, straight, or combination of them



*Ref.: National Programme on Technology Enhanced Learning (NPTEL)

BASIC CONCEPTS

The objectives of providing camber are:

- Surface protection especially for gravel and bituminous roads
- Sub-grade protection by proper drainage
- Quick drying of pavement which in turn increases safety

Too steep slope is undesirable for it will erode the surface. Camber is measured in *1 m n* or *n%* (Eg. 1 m 50 or 2%) and the value depends on the type of pavement surface. The values suggested for various categories of pavement is given in below table.

Surface type	Heavy rain	Light rain
Concrete/Bituminous	2 %	1.7 %
Gravel/WBM	3 %	2.5 %
Earthen	4 %	3.0 %

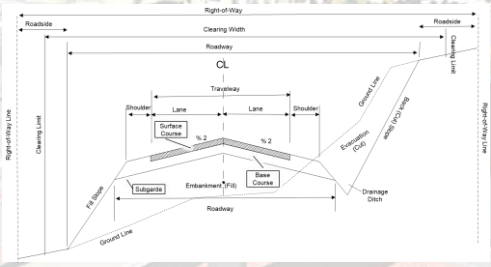
*Ref.: National Programme on Technology Enhanced Learning (NPTEL)

BASIC CONCEPTS

Cut: The part of earth above the grade line of a road that has to be cut. The cut part is used as fill or sent to storage.

Fill: The part of earth under the grade line of a road that has to be filled. The fill part is provided from the cut part road or from the quarry.

Drainage: To draw off the ground water without harming the road facilities; the act of draining.



BASIC CONCEPTS

Earthwork, grading: An embankment or other construction made of earth; especially, one used as a field fortification; the operations connected with excavations and embankments of earth.

Formation: The surface of the base built by the earthwork which is prepared for the superstructure.

Grading-Finishing: Levelling the road in order to give the longitudinial and latidunnial grades at the end of the earthwork.

Sub-grade: The level layer of rock or earth upon which the foundation of a road or railway is laid; the ground beneath a roadway or pavement

Pavement: The combination of subbase, base, base course and surface course placed on a subgrade to support the traffic load and distribute it to the road bed.

Base course: A layer of selected material of planned thickness, constructed on the subgrade or subbase for the purpose of serving one or more functions such as distributing load, providing drainage, minimizing frost action, etc.

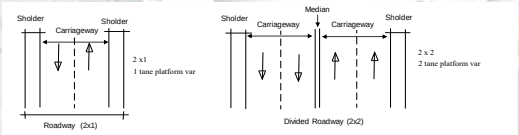
Sub-base course: For the support of paving materials, a layer of granular material applied to the subgrade in a predetermined depth; provides

Shoulder: A portion of the roadway contiguous with the traveled way for accommodation of stopped vehicles, emergency use, and lateral support of the subbase, base, and surface courses.

BASIC CONCEPTS

Roadway (Platform): A road (especially that part of a road) over which vehicles travel;Carriageway (travelway); One of the two sides of a motorway where traffic travels in one direction only usually in two or three lanes.
Surfacing, pavement: The hard layered structure that forms a road carriageway, airfield runway, vehicle park, or other paved areas.

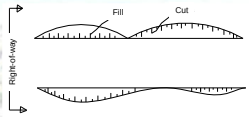
Median, central reserve: A median is the portion of a roadway separating opposite directions of the traffic flows.





BASIC CONCEPTS

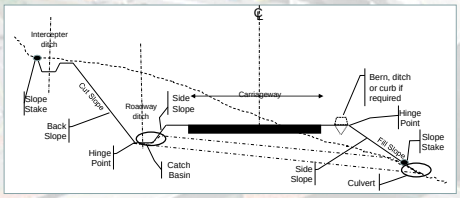
Right-of-way width: The legal right of someone to pass over another's land, acquired by grant or by long usage; the path or road used by this right.



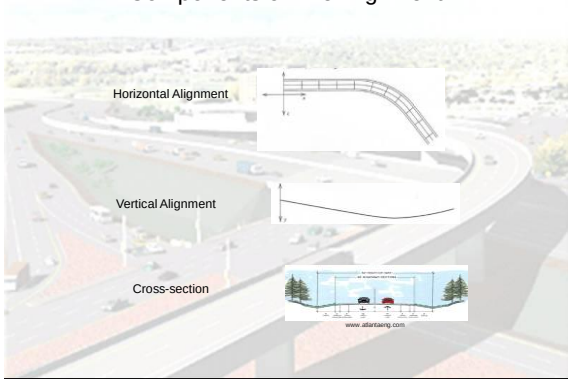
Ditch: A small artificial channel cut through earth or rock to carry water for irrigation or drainage. A long narrow cut made in the earth to bury pipeline, cable, or similar installations.
Slope, crossfall: Surface cross slopes are necessary on all components of the crosssection to facilitate drainage.
Climbing lane: A passing lane added on an upgrade to allow traffic to pass heavy vehicles whose speeds are reduced.
Refuge: A place that provides shelter or protection; elevated part of the road to seperate the traffic flows and provide safe walking space for pedestrains
Barrier : Separates the traffic going in different directions using by guardrail or the other separators
Curb: Curbs, berms, and edging are roadside elements, usually constructed of granite or extruded bituminous concrete used to define the pavement edge and to control drainage.

BASIC CONCEPTS

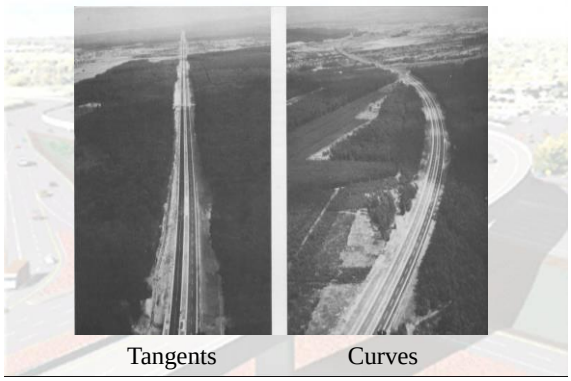
Slope stake: Slope staking is a trial and error process of finding the point where the slope intercepts the natural ground. This point is referred to as the "catch point" or "slope catch."
Intercepting ditch: An interceptor ditch is a small ditch or channel constructed to intercept and convey water to an area where it can be safely discharged.
Side slope: A long narrow hole that is dug along a road, field, etc., and used to hold or move water; a long narrow hole that is dug along a road, field, etc., and used to hold or move water.



Components of The Alignment



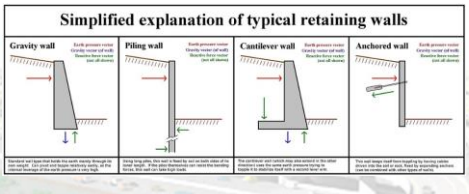
Horizontal Alignment

















BASIC CONCEPTS

Type of Highway

Divided road, dual carriageway: A highway divided down the middle by a barrier that separates traffic going in different directions with dual carriageway.

Undivided Road : Bidirectional road with only carriageway

Full access control: A roadway is only accessed via entrance and exit ramps, which allow traffic to enter and exit the main lanes of the roadway at (nearly) full speed. All cross-traffic is grade-separated from the access-controlled roadway.

Partial access control: Direct access to and from adjacent private properties is prohibited or extremely limited. A partially-controlled-access highway can (but does not always) have at-grade intersections with cross-roads, but the frequency of such intersections is usually limited. Often, a partially-controlled-access highway has grade-separated interchanges, rather than at-grade intersections, with major crossing highways.

Freeway, Motorway: A multilane, divided highway with a minimum of two lanes for the exclusive use of traffic in each direction and full control of access without traffic interruption.

BASIC CONCEPTS

Codes of State Highway

